

Technology Stack

Date	27 June 2026
Team ID	XXXXXX
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side (e.g., React, Angular, HTML/CSS)	Streamlit, HTML, CSS	Streamlit provides a responsive web interface for emotion analysis. HTML and CSS enhance the application's appearance and improve the user experience.

2	Backend / Server-Side (e.g., Python/Django, Java)	Python	Python is used to implement text preprocessing, emotion detection, AI integration, and application logic because of its powerful AI and machine learning ecosystem.
3	Database / Data Storage (e.g., MySQL, MongoDB, PostgreSQL)	CSV Files, Pickle (.pkl), NumPy (.npy)	CSV files store emotion records and session history. Pickle files store the tokenizer and label encoder, while NumPy files store padded sequences for efficient model processing.
4	Cloud / Hosting / Deployment (e.g., AWS, Azure, Docker)	Local Deployment using Streamlit	The application is currently deployed locally for development, testing, and demonstration. It can be migrated to cloud platforms such as Streamlit Cloud, Azure, or AWS in future versions.
5	Version Control & CI/CD (e.g., GitHub, GitLab, Jenkins)	Git, GitHub	Git and GitHub are used for source code management, version tracking, collaboration among team members, and project backup.

6	Third-Party APIs / Other Tools (e.g., Google APIs, ML Libraries)	Google Gemini API, TensorFlow, Hugging Face Transformers, NLTK, Pandas, NumPy, Plotly, Joblib	Google Gemini API generates personalized learning guidance. TensorFlow powers the BiLSTM model, Hugging Face Transformers support the BERT model, NLTK performs text preprocessing, Pandas and NumPy handle data processing, Plotly creates interactive visualizations, and Joblib is used for model serialization.
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